

Schrödinger Bridge & Stochastic Optimal Control on Discrete Spaces

► Continuous-time Markov chains (CTMCs)

Let $\mathcal{X} = [N]^D$ and $(X_t)_{t \in [0,1]}$ be a CTMC with **transition rate** $r_t(y, x) = \lim_{h \rightarrow 0} \frac{1}{h} (\Pr(X_{t+h} = y | X_t = x) - 1_{y=x})$.

Aim: learn to sample from a target distribution $\nu \propto e^{-U}$ on $\mathcal{X}$ by learning a CTMC to transport a simple prior distribution μ to ν .

► Schrödinger Bridge (SB) problem

Let $r_t(y, x)$ and $u_t(y, x)$ denote the reference and controlled transition rates, and write p^r and p^u for their induced **path measures**. Assume they share the same initial distribution $p_0^r = p_0^u = \mu$. Consider the following **SB problem**:

$$\min_{u \text{ s.t. } p_1^u = \nu} \left\{ \text{KL}(p^u \| p^r) = \mathbb{E}_{X \sim p^u} \int_0^1 \sum_{y \neq X_t} \left(u_t \log \frac{u_t}{r_t} + r_t - u_t \right) (y, X_t) dt \right\}.$$

► Optimal transition rate & SB potentials

The optimum factorizes through a forward–backward pair $(\varphi_t, \hat{\varphi}_t)$:

$$u_t^*(y, x) = \frac{\varphi_t(y)}{\varphi_t(x)} r_t(y, x), \quad \forall y \neq x,$$

$$\varphi_s(x) = \sum_y p_{t|s}^r(y|x) \varphi_t(y), \quad \hat{\varphi}_t(x) = \sum_y p_{t|s}^r(x|y) \hat{\varphi}_s(y), \quad \forall 0 \leq s < t \leq 1,$$

with $p_t^* = \varphi_t \hat{\varphi}_t$ and the SB-coupling identities $\frac{p_{t|s}^*(x|y)}{p_{t|s}^r(x|y)} = \frac{\varphi_t(x)}{\varphi_s(y)}, \frac{p_{s|t}^*(y|x)}{p_{s|t}^r(x|y)} = \frac{\hat{\varphi}_s(y)}{\hat{\varphi}_t(x)}$.

► SOC reformulation & SB-SOC equivalence

$$\min_{u \text{ s.t. } p_0^u = \mu} \left\{ \text{KL}(p^u \| p^r) + \mathbb{E}_{X \sim p^u} g(X_1) \right\}.$$

Choosing $g \leftarrow \log \frac{\hat{\varphi}_1}{\nu}$ yields the same u^* as SB.

The terminal-marginal constraint of SB becomes a tractable terminal cost in SOC, opening the door to AM-style learning.

► Reference dynamics & cyclic group structure

We use the **uniform reference rate**

$$r_t(y, x) = \frac{2}{N} 1_{d_H(x,y)=1}, \quad x \neq y \in [N]^D,$$

and view $\mathcal{X} = [N]^D$ as the **cyclic group** $\mathbb{Z}_N^D$. The reference transition is **additive**: $p_{1|t}^r(y|x) = q_t(y - x)$, mirroring the additive Brownian noise in continuous AM.

Denoising Matching vs. Adjoint Matching

DM holds for general reference dynamics, while **AM** requires **additive** noise.

	Continuous state space $\mathcal{X} = \mathbb{R}^D$	Discrete state space $\mathcal{X} = \mathbb{Z}_N^D$
Ref. dyn. p^r	$dX_t = b_t(X_t)dt + \sigma_t dW_t, X_0 \sim \mu$	CTMC (X_t) with rate $(r_t), X_0 \sim \mu$
Ctrl. dyn. p^u	$dX_t = (b_t + \sigma_t u_t)(X_t)dt + \sigma_t dW_t, X_0 \sim \mu$	CTMC (X_t) with rate $(u_t), X_0 \sim \mu$
SB Prob.	$\min_{u \text{ s.t. } X_1 \sim \nu} \mathbb{E}_{X \sim p^u} \left[\int_0^1 \frac{1}{2} \ u_t(X_t)\ ^2 dt \right]$	$\min_{u \text{ s.t. } X_1 \sim \nu} \mathbb{E}_{X \sim p^u} \left[\int_0^1 \sum_{y \neq X_t} (u_t \log \frac{u_t}{r_t} + r_t - u_t)(y, X_t) dt \right]$
SOC Prob.	$\min_u \mathbb{E}_{X \sim p^u} \left[\int_0^1 \frac{1}{2} \ u_t(X_t)\ ^2 dt + \log \frac{\hat{\varphi}_1}{\nu}(X_1) \right]$	$\min_u \mathbb{E}_{X \sim p^u} \left[\int_0^1 \sum_{y \neq X_t} (u_t \log \frac{u_t}{r_t} + r_t - u_t)(y, X_t) dt + \log \frac{\hat{\varphi}_1}{\nu}(X_1) \right]$
Opt. Ctrl.	$u_t^*(x) = \sigma_t \nabla \log \varphi_t(x)$	$u_t^*(y, x) = r_t(y, x) \frac{\varphi_t(y)}{\varphi_t(x)}$
Corrector	$\nabla \log \hat{\varphi}_1(x)$	$\frac{\hat{\varphi}_1(y)}{\hat{\varphi}_1(x)}$
Example of Additive Noise	$b_t \equiv 0 \Rightarrow [p_{1 t}^r(x_1 x) = q_t(x_1 - x)]$ $q_t(\varepsilon) = \mathcal{N}(\varepsilon; 0, \tilde{\sigma}_t^2 I)$	$r_t(y, x) = \frac{2}{N} 1_{d_H(x,y)=1} \Rightarrow [p_{1 t}^r(x_1 x) = q_t(x_1 - x)]$ $q_t(\varepsilon) = A(t, 1)^{d_H(\varepsilon, 0)} B(t, 1)^{D-d_H(\varepsilon, 0)}$
Denoising Matching	$\nabla \log \varphi_t(x) = \mathbb{E}_{p_{t t}^r(x_1 x)} \nabla_x \log p_{t t}^r(x_1 x)$ $\nabla \log \hat{\varphi}_1(x) = \mathbb{E}_{p_{0 1}^r(x_0 x)} \nabla_x \log p_{1 0}^r(x x_0)$	$\frac{\varphi_t(y)}{\varphi_t(x)} = \mathbb{E}_{p_{t t}^r(x_1 x)} \frac{p_{t t}^r(x_1 y)}{p_{t t}^r(x_1 x)}$ $\frac{\hat{\varphi}_1(y)}{\hat{\varphi}_1(x)} = \mathbb{E}_{p_{0 1}^r(x_0 x)} \frac{p_{1 0}^r(y x_0)}{p_{1 0}^r(x x_0)}$
Adjoint Matching	$\nabla \log \varphi_t(x) = \mathbb{E}_{p_{t t}^r(x_1 x)} \nabla \log \varphi_1(x_1)$ $\nabla \log \hat{\varphi}_1(x) = \mathbb{E}_{p_{0 1}^r(x_0 x)} \nabla \log \hat{\varphi}_0(x_0)$	$\frac{\varphi_t(y)}{\varphi_t(x)} = \mathbb{E}_{p_{t t}^r(x_1 x)} \frac{\varphi_1(y+x_1-x)}{\varphi_1(x_1)}$ $\frac{\hat{\varphi}_1(y)}{\hat{\varphi}_1(x)} = \mathbb{E}_{p_{0 1}^r(x_0 x)} \frac{\hat{\varphi}_0(y+x_0-x)}{\hat{\varphi}_0(x_0)}$

Discrete ASBS: Alternating Controller–Corrector Updates

Define controller $\Phi_t^*(x)_{d,n} = \frac{\varphi_t(x^{d \leftarrow n})}{\varphi_t(x)}$ and corrector $\hat{\Phi}^*(x)_{d,n} = \frac{\hat{\varphi}_1(x^{d \leftarrow n})}{\hat{\varphi}_1(x)}$. At stage k :

$$\Phi^{(k)} := \operatorname{argmin}_{\Phi} \mathbb{E}_t w_t \mathbb{E}_{p_{0,1}^{\text{sg}(r\Phi)}(x_0, x_1)} \sum_{d, n \neq x^d} D_f \left(\frac{\varphi_1(x_1^{d \leftarrow x_1^d + n - x^d})}{\varphi_1(x_1)} \left\| \Phi_t(x)_{d,n} \right\| \right),$$

$$\hat{\Phi}^{(k)} := \operatorname{argmin}_{\hat{\Phi}} \mathbb{E}_{p_{0,1}^{\text{sg}(r\Phi^{(k)})}(x_0, x_1)} \sum_{d, n \neq x_1^d} D_f \left(\frac{\hat{\varphi}_0(x_0^{d \leftarrow x_0^d + n - x_1^d})}{\hat{\varphi}_0(x_0)} \left\| \hat{\Phi}(x_1)_{d,n} \right\| \right),$$

$$\text{or } \hat{\Phi}^{(k)} := \operatorname{argmin}_{\hat{\Phi}} \mathbb{E}_t w_t \mathbb{E}_{p_{0,1}^{\text{sg}(r\Phi^{(k)})}(x_0, x_1)} \sum_{d, n \neq x_1^d} D_f \left(\frac{p_{t|t}^r(x_1^{d \leftarrow n}|x)}{p_{t|t}^r(x_1|x)} \left\| \hat{\Phi}(x_1)_{d,n} \right\| \right).$$

The **AM** target uses the cyclic-group shift $x_1^d + n - x^d$, well-defined precisely because we identify $[N]^D$ with $\mathbb{Z}_N^D$.

◆ Convergence guarantee (fixed-point view)

The controller update is a **forward half-bridge** projection; the corrector update is a **backward half-bridge** projection. By **iterative proportional fitting**, alternating updates converge to the SB optimum p^* .

Why Adjoint Matching Wins on Additive References

◆ Continuous AM/DM analogy

For continuous random vectors $x, y \sim p(x, y)$,

$$\underbrace{\mathbb{E}_{p(x|y)} \nabla_x \log p(x)}_{\text{additive } p(y|x): \text{ target matching}} = \nabla_y \log p(y) = \underbrace{\mathbb{E}_{p(x|y)} \nabla_y \log p(y|x)}_{\text{general } p(y|x): \text{ denoising matching}}.$$

TM regresses on *target-side* scores; **DM** on *kernel-side* scores. The same dichotomy carries to discrete AM: explicit mutual supervision between Φ and $\hat{\Phi}$, no singularity at small noise.

◆ Unified view

Adjoint matching: a fixed-point iteration driven by a target matching objective converging to the optimal p^ .*

Experiments

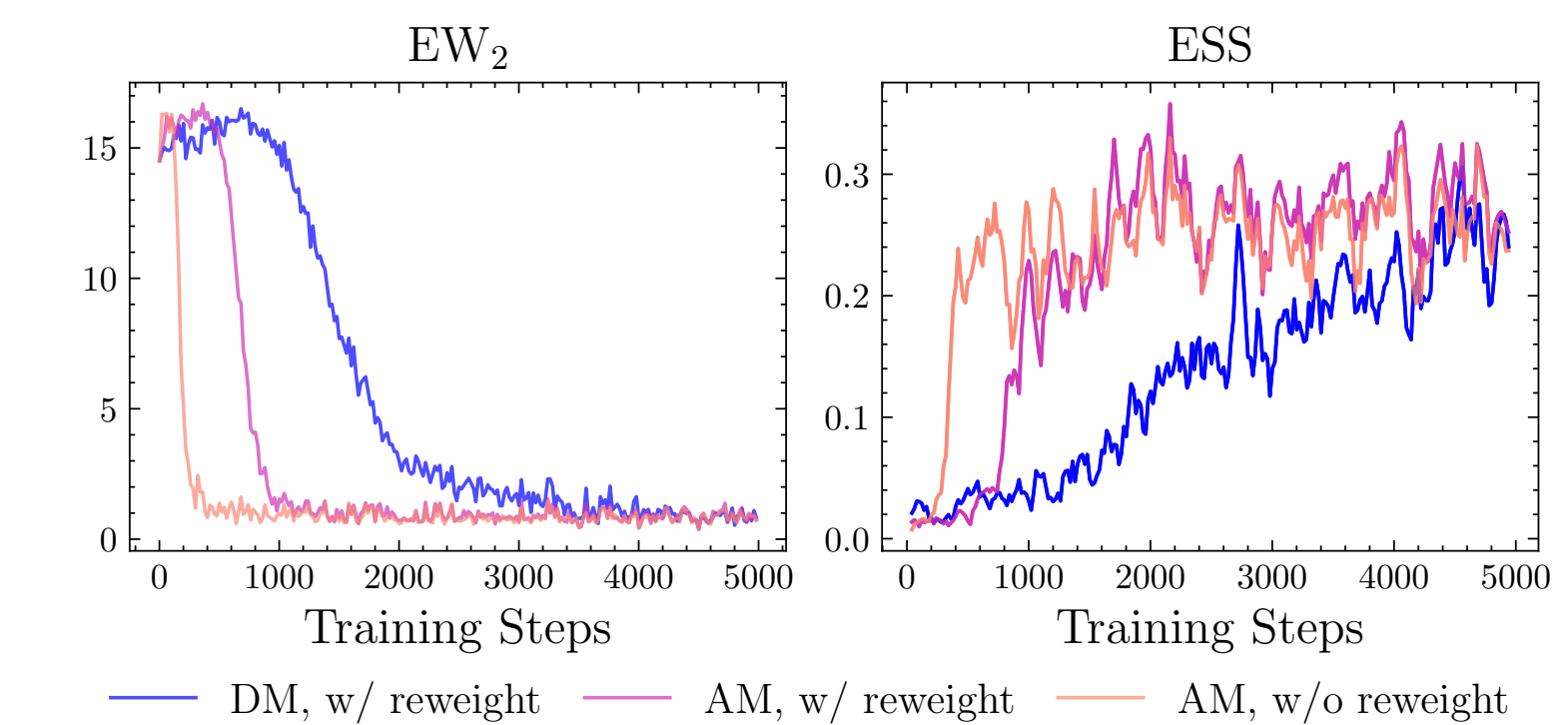
♣ Learning to sample from Ising models ($L = 24$)

Inv. Temp.			$\beta_{\text{high}} = 0.28$			$\beta_{\text{critical}} = 0.4407$			$\beta_{\text{low}} = 0.6$			
Type	Metrics ↓	Steps (×1e3)	Runtime (h)	ΔMag.	ΔCorr.	EW ₂	ΔMag.	ΔCorr.	EW ₂	ΔMag.	ΔCorr.	EW ₂
Uniform	DASBS	3.75	0.5	<u>2.7e−3</u>	<u>1.8e−3</u>	<u>8.6</u>	5.7e−2	4.7e−2	12.1	2.0e−2	1.8e−3	2.0
	LEAPS	30	8.4	1.8e−3	9.2e−4	3.1	<u>5.9e−2</u>	<u>2.8e−1</u>	<u>96.5</u>	<u>3.0e−2</u>	<u>5.5e−1</u>	<u>176.6</u>
	UDNS [†]	50	11.9	9.0e−3	8.7e−3	23.6	−	−	−	−	−	−
	DFNS [‡]	50	<u>2.1</u>	9.3e−1	8.0e−1	661.6	−	−	−	−	−	−
Masked	MDNS [†]	50	16.8	3.9e−3	7.4e−4	0.1	1.1e−2	5.6e−3	5.1	9.0e−3	4.7e−3	5.3
MCMC	MH	−	−	8.9e−4	2.9e−4	1.2	2.5e−2	3.7e−3	293.3	4.0e−2	6.6e−4	109.9

Best/2nd among *uniform-based* samplers on one A6000. †: warm-up. ‡: failed at $\beta_{\text{critical}}, \beta_{\text{low}}$.

- DASBS reaches competitive quality at the **lowest training cost** among uniform-based samplers, thanks to the *first-order* discrete-score oracle + memory-light per-step loss.

♣ Ablation study of AM v.s. DM on Potts model



- **AM** converges much faster than **DM**; reweighting can hurt (small ESS).